

COURIER MANAGEMENT SYSTEM

Developed By (Team Members)

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Abstract

A Courier Management System (CMS) is a software application used to manage courier services efficiently. It helps organizations track parcels, manage customer information, schedule deliveries, and generate reports.

The system allows customers to register their courier details and track shipments using a unique tracking ID. It reduces manual work and increases efficiency in courier service operations.

Introduction

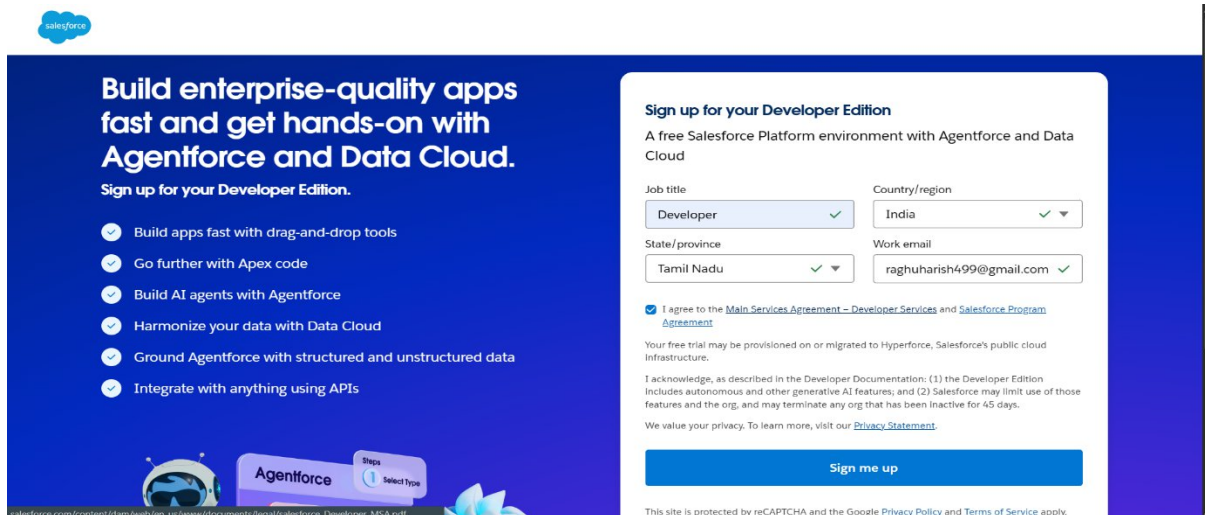
1.1 Project Overview

The main purpose of the **Courier Management System** is to automate and centralize courier and logistics operations using the Salesforce platform. The system aims to replace manual and fragmented processes with an integrated digital platform that improves efficiency and accuracy in courier services.

The CMS helps organizations manage shipments, customers, delivery agents, and billing information from a single system. It provides real-time tracking of parcels and ensures that delivery updates are recorded accurately. By automating tasks such as delivery agent assignment and invoice generation, the system reduces human errors and speeds up the overall workflow.

Another important purpose of this system is to improve communication between courier companies and customers. Customers can track their shipments and receive timely updates about delivery status, which increases trust and satisfaction. The system also provides analytical reports and dashboards that help managers evaluate operational performance and make informed decisions.

In summary, the Courier Management System is designed to improve service efficiency, enhance customer experience, streamline courier operations, and support business growth through automation and real-time information management.



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1.2 Purpose of the Project

The purpose of the Courier Management System is to automate courier management operations and replace manual processes with a digital platform.

Objectives

- Automate shipment booking
- Provide real-time parcel tracking
- Assign delivery agents automatically
- Generate invoices automatically
- Improve customer communication
- Provide analytics and reports

2. Problem Statement

<i>Problem ID</i>	<i>Customer Type</i>	<i>Problem</i>	<i>Reason</i>	<i>Impact</i>
<i>PS-1</i>	<i>Branch Manager</i>	<i>Difficulty managing shipments</i>	<i>Manual tracking system</i>	<i>Stress and operational delay</i>
<i>PS-2</i>	<i>Customer</i>	<i>Cannot track parcels easily</i>	<i>No centralized system</i>	<i>Uncertainty about delivery</i>

2.1 Functional Requirements Table Explanation

FR No Requirement

FR-1 Customer Management

FR-2 Shipment Booking

FR-3 Delivery Agent Assignment

FR-4 Real-time Status Updates

FR-5 Invoice Generation

FR-6 Reports & Dashboards

Explanation

FR-1 Customer Management

This feature allows the system to store and manage customer information such as name, email, phone number, and address. It helps the courier company maintain accurate customer records.

FR-2 Shipment Booking

This module enables users to create shipment records when a customer sends a parcel. It stores details like tracking number, source address, destination address, weight, and shipment type.

FR-3 Delivery Agent Assignment

This function assigns a delivery agent to a particular shipment. It ensures that parcels are delivered by the appropriate agent and helps manage delivery responsibilities.

FR-4 Real-time Status Updates

The system updates shipment status such as *Booked*, *In Transit*, *Out for Delivery*, *Delivered*, or *Cancelled*. Customers and managers can track the parcel status instantly.

FR-5 Invoice Generation

This feature automatically generates invoices for shipments. It records the shipment price, invoice date, and payment status.

FR-6 Reports and Dashboards

This module provides analytical reports and dashboards that help managers monitor shipment performance, delivery efficiency, and revenue.

2.2 Non-Functional Requirements Table Explanation

NFR	Description
Usability	Easy Lightning interface
Security	Role-based access control
Reliability	Error-free automation
Performance	Fast tracking updates
Availability	24/7 cloud access
Scalability	Supports growing branches

Explanation

Usability

The system provides a user-friendly interface using Salesforce Lightning, making it easy for users to operate the application.

Security

Role-based access control ensures that only authorized users can access specific data or functions.

Reliability

Automation features reduce manual errors and ensure accurate system operations.

Performance

The system processes shipment tracking and updates quickly to improve efficiency.

Availability

Since Salesforce is cloud-based, the system can be accessed anytime and anywhere.

Scalability

The system can support multiple courier branches as the business grows.

3. PROJECT DESIGN PHASE

3.1 Problem Solution Fit

Problems	Solutions
Manual shipment tracking	Automated tracking
Delayed updates	Real-time updates

Problems

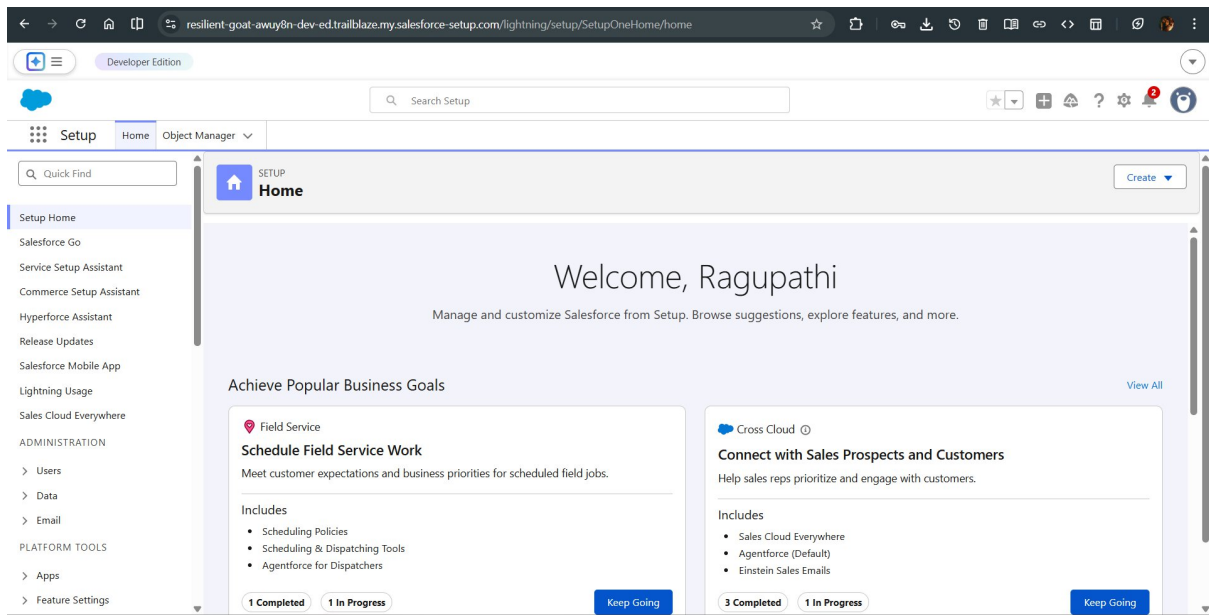
Scattered records

Customer confusion

Solutions

Centralized database

Shipment tracking system



3.2 Proposed Solution

The proposed system automates courier operations using Salesforce.

Main features:

- Customer management
- Shipment tracking
- Delivery agent management
- Invoice generation
- Reporting dashboard

3.3 System Architecture

Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object

- Delivery Update Object
- Invoice Object
- Automation Flow
- Reports and Dashboards

4. PROJECT PLANNING AND SCHEDULING

Project planning is the process of organizing tasks, resources, and timelines to successfully complete the Courier Management System. The project follows the Agile development methodology, where the work is divided into small phases called sprints.

Each sprint focuses on completing specific tasks such as system setup, data modeling, automation, security configuration, and reporting. This approach helps the development team complete the project step by step and track progress easily.

The planning stage includes creating a product backlog, defining user stories, estimating story points, and assigning tasks to team members. This ensures that all project activities are completed within the planned schedule.

Sprint Plan

Sprint	Task
Sprint 1	Salesforce developer account setup
Sprint 2	Creating objects and data modeling
Sprint 3	Implementing automation and validation rules
Sprint 4	Configuring security roles and profiles
Sprint 5	Creating reports
Sprint 6	Developing dashboards

Through proper project planning, the Courier Management System is developed efficiently with clear task distribution and timely completion.

5. Objects in Courier Management System

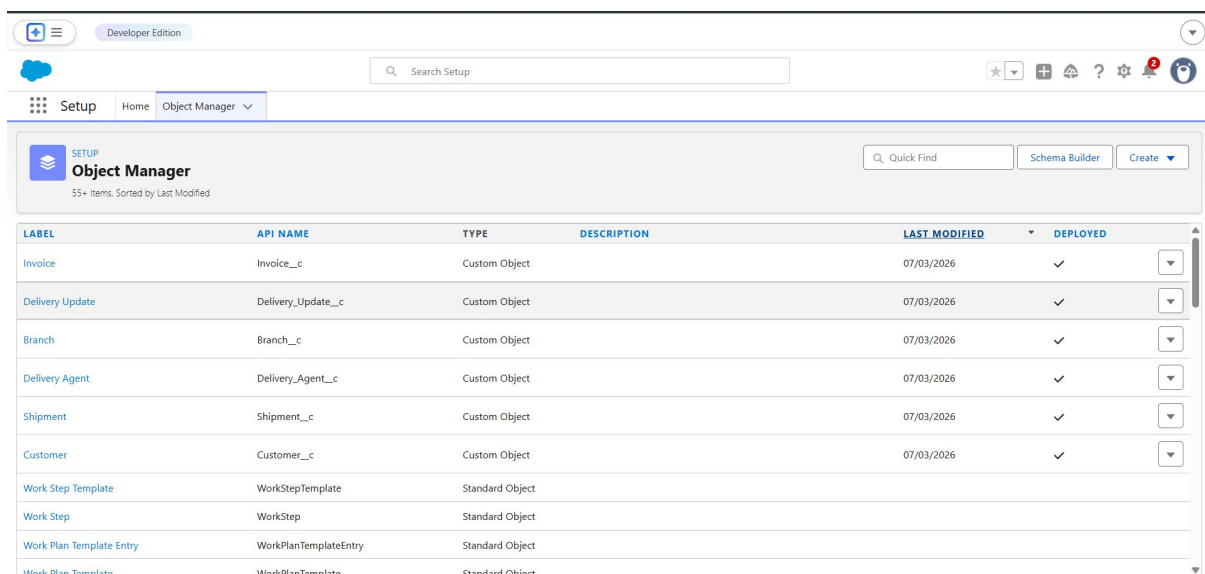
In Salesforce, an **object** is similar to a database table used to store information in the form of records and fields. In the Courier Management System, custom objects are created to manage courier operations such as customers, shipments, agents, and invoices.

The main objects used in this project are:

Object Name Description

Customer	Stores customer details such as name, phone number, email, and address.
Shipment	Stores parcel information including tracking number, source address, destination address, and shipment status.
Delivery Agent	Maintains delivery agent details such as name, contact number, and assigned branch.
Branch	Stores information about courier branch offices like branch name, city, and manager.
Delivery Update	Records shipment status updates such as <i>In Transit</i> , <i>Out for Delivery</i> , <i>Delivered</i> , or <i>Delayed</i> .
Invoice	Stores billing details including shipment amount, invoice date, and payment status.

These objects are connected using **relationships** so that the system can track shipments, assign delivery agents, and generate invoices efficiently.



The screenshot shows the Salesforce Setup interface, specifically the Object Manager. The top navigation bar includes 'Setup', 'Home', and 'Object Manager'. The main content area displays a table of objects with columns for Label, API Name, Type, Description, Last Modified, and Deployed. The objects listed are Invoice, Delivery Update, Branch, Delivery Agent, Shipment, Customer, Work Step Template, Work Step, Work Plan Template Entry, and Work Plan Template. The first six objects are Custom Objects, while the last four are Standard Objects. All Custom Objects are marked as Deployed.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Invoice	Invoice__c	Custom Object		07/03/2026	✓
Delivery Update	Delivery_Update__c	Custom Object		07/03/2026	✓
Branch	Branch__c	Custom Object		07/03/2026	✓
Delivery Agent	Delivery_Agent__c	Custom Object		07/03/2026	✓
Shipment	Shipment__c	Custom Object		07/03/2026	✓
Customer	Customer__c	Custom Object		07/03/2026	✓
Work Step Template	WorkStepTemplate	Standard Object			
Work Step	WorkStep	Standard Object			
Work Plan Template Entry	WorkPlanTemplateEntry	Standard Object			
Work Plan Template	WorkPlanTemplate	Standard Object			

6. Creating Form Tabs (Step by Step)

Tabs in Salesforce help users easily access object records such as **Customer**, **Shipment**, **Delivery Agent**, **Branch**, **Delivery Update**, and **Invoice**.

Step 1: Open Setup

1. Login to your **Salesforce account**.

2. Click the **Gear Icon (⚙)** on the top right.
3. Select **Setup**.

Step 2: Search Tabs

1. In the **Quick Find** search box, type **Tabs**.
2. Click **Tabs** under **User Interface**.

Step 3: Create New Tab

1. Scroll down to **Custom Object Tabs**.
2. Click **New**.

Step 4: Select Object

1. In the **Object** dropdown list, select the object.
Example:
 - o Customer
 - o Shipment
 - o Delivery Agent
 - o Branch
 - o Delivery Update
 - o Invoice

Step 5: Select Tab Style

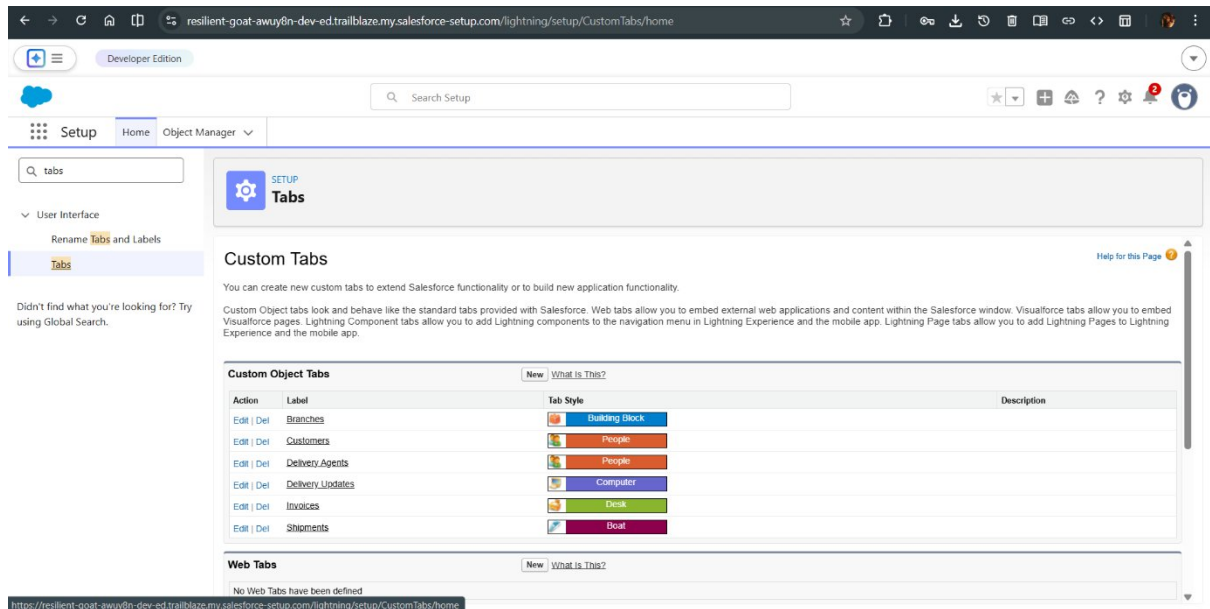
1. Choose any **Tab Style (Icon)** for the object.
2. Click **Next**.

Step 6: Set Profile Visibility

1. Select the profiles that can see the tab.
2. Usually select **System Administrator**.
3. Click **Next**.

Step 7: Add to Custom Apps

1. Select the app where the tab should appear.
2. Example: **Courier Management System App**.



Open App Manager

1. In the **Quick Find** search box, type **App Manager**.
2. Click **App Manager**.

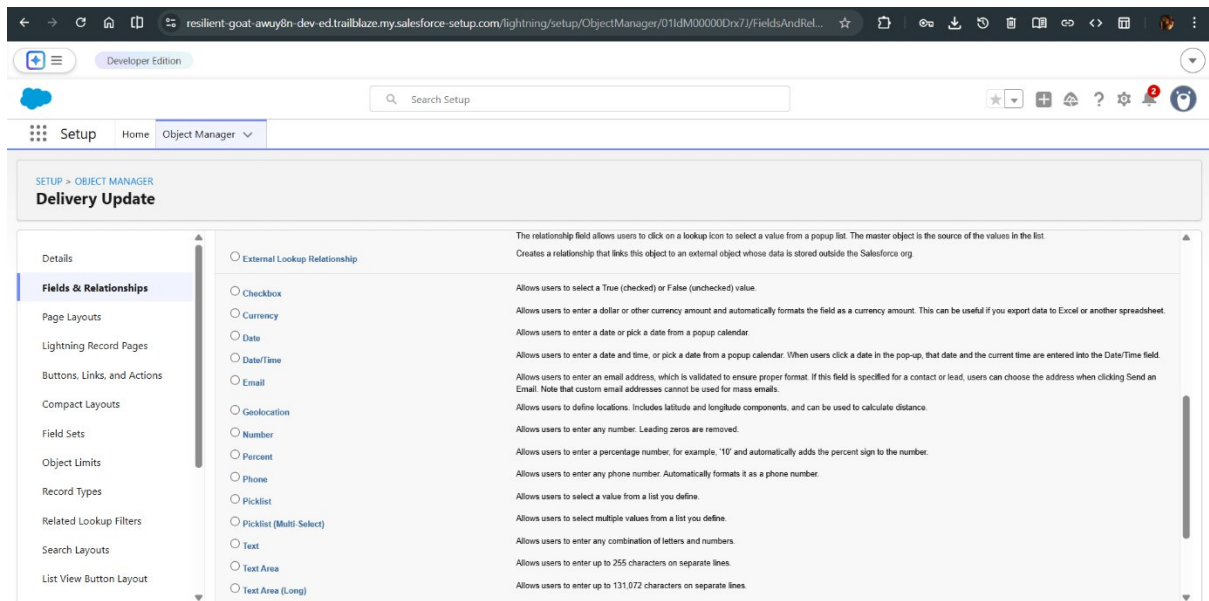
Create New Lightning App

1. Click **New Lightning App**.
2. Enter the following details:

Field	Value
-------	-------

App Name	Courier Management System
----------	---------------------------

Description	Courier management application
-------------	--------------------------------



7. Profiles in Courier Management System

In Salesforce, a **Profile** is a collection of permissions and settings that control what a user can access and perform in the system. Profiles define the level of access to objects, fields, apps, and system features.

In the **Courier Management System**, different profiles are created to manage user permissions based on their roles and responsibilities.

Profiles help ensure **data security and proper access control** in the system.

Profiles Created in the Project

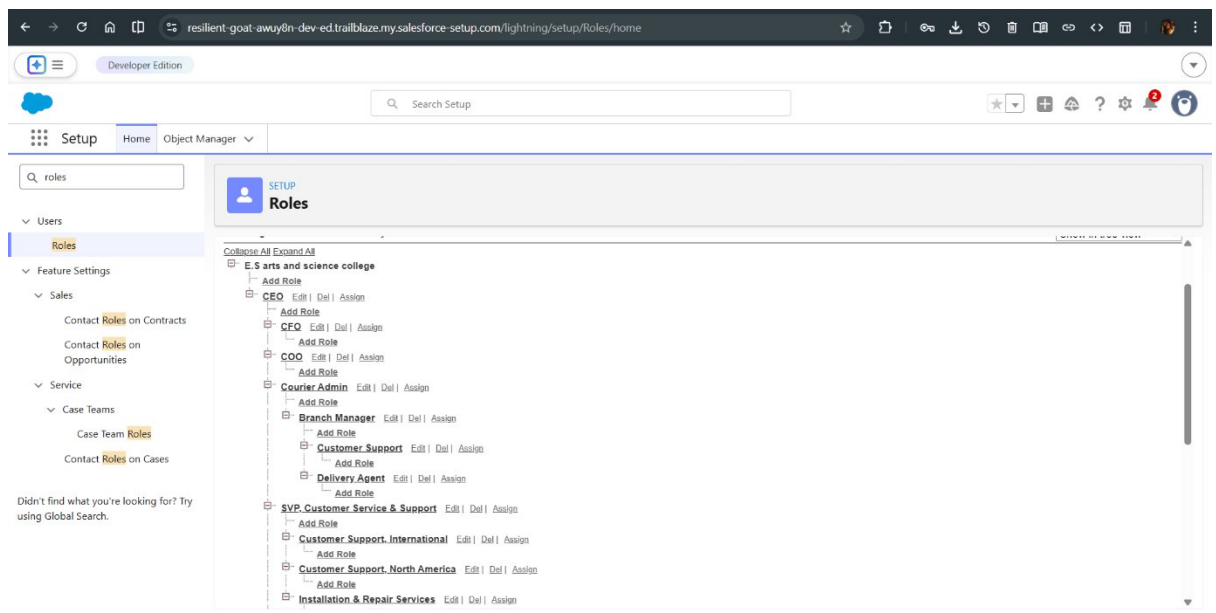
Profile Name Purpose

Courier Admin	Full access to manage the entire system
Branch Manager	Manage shipments, agents, and branch operations
Delivery Agent	Update shipment delivery status
Customer Support	Handle customer queries and track shipments

Steps to Create a Profile

1. Login to **Salesforce**.

- ## 8. Creating (Courier Management System)



In Salesforce, an **App** is a collection of tabs and objects that work together to perform specific tasks. Creating an app helps users easily access different modules such as customers, shipments, delivery agents, and invoices in one place.

In this project, a **Lightning App** called "**Courier Management System**" is created to manage courier operations efficiently.

Open App Manager

1. In the **Quick Find** box, type **App Manager**.
2. Click **App Manager**.

Step 1: Create New Lightning App

1. Click **New Lightning App**.
2. Enter the following details:

Field	Value
--------------	--------------

App Name	Courier Management System
----------	---------------------------

Description	Courier Management Application
-------------	--------------------------------

3. Click **Next**.

Step 2: Configure App Options

1. Keep the default settings.
2. Click **Next**.

Step 3: Select Utility Items

1. Leave the default options.
2. Click **Next**.

Step 4: Add Navigation Items

Add the following items to the app:

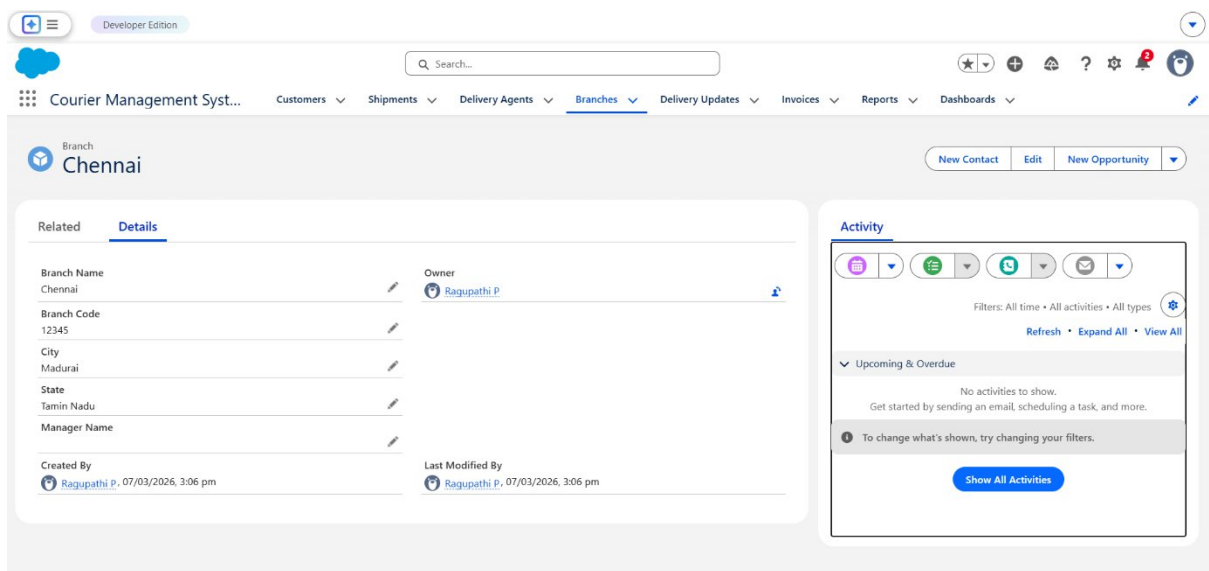
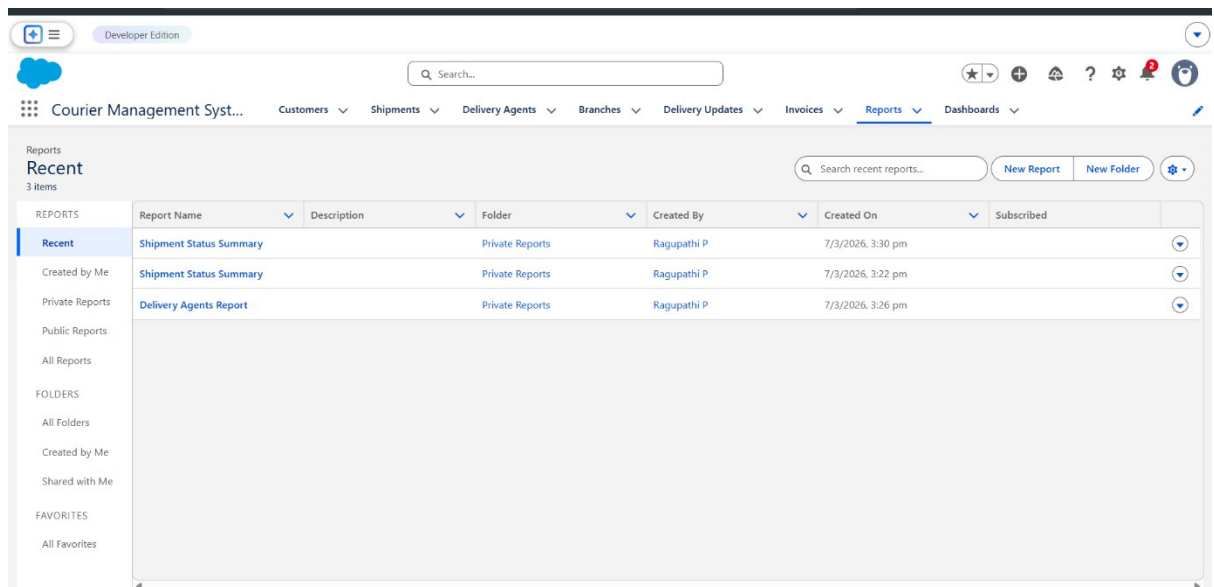
- Customer
- Shipment
- Delivery Agent

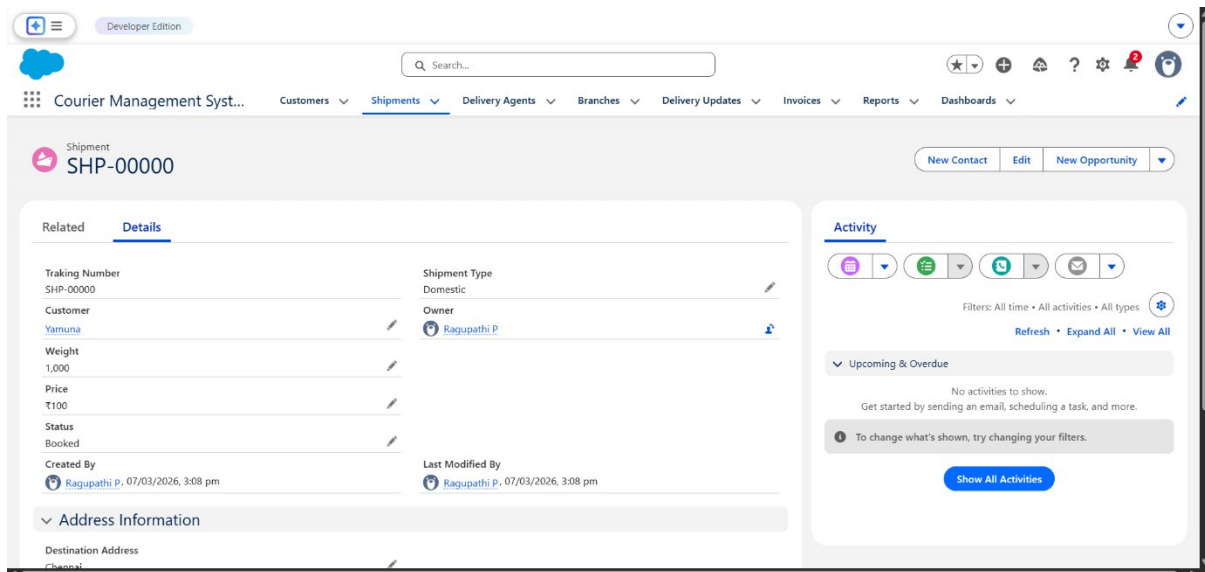
- Branch
- Delivery Update
- Invoice
- Reports
- Dashboards

Click **Next**.

Step 5: Assign User Profiles

1. Select **System Administrator** profile.
2. Click **Save & Finish**.





9.Delivery Agent

In Salesforce, the **Delivery Agent object** is created to store and manage information about the delivery personnel responsible for delivering shipments. This object helps the courier company track agent details, assign shipments, and monitor delivery status.

The Delivery Agent object is created in **Object Manager** as a **custom object**. It stores important information such as the agent's name, email, phone number, branch, and working status.

This object is connected with other objects like **Shipment** and **Branch** to manage courier operations efficiently.

Fields in Delivery Agent Object

Field Name Data Type Description

Agent Name	Text	Name of the delivery agent
Email	Email	Agent email address
Phone	Phone	Contact number of the agent
Branch	Lookup	Branch where the agent works
Status	Picklist	Agent status (Active, Inactive, On Leave)
Joining Date	Date	Date when the agent joined

Purpose of Delivery Agent Object

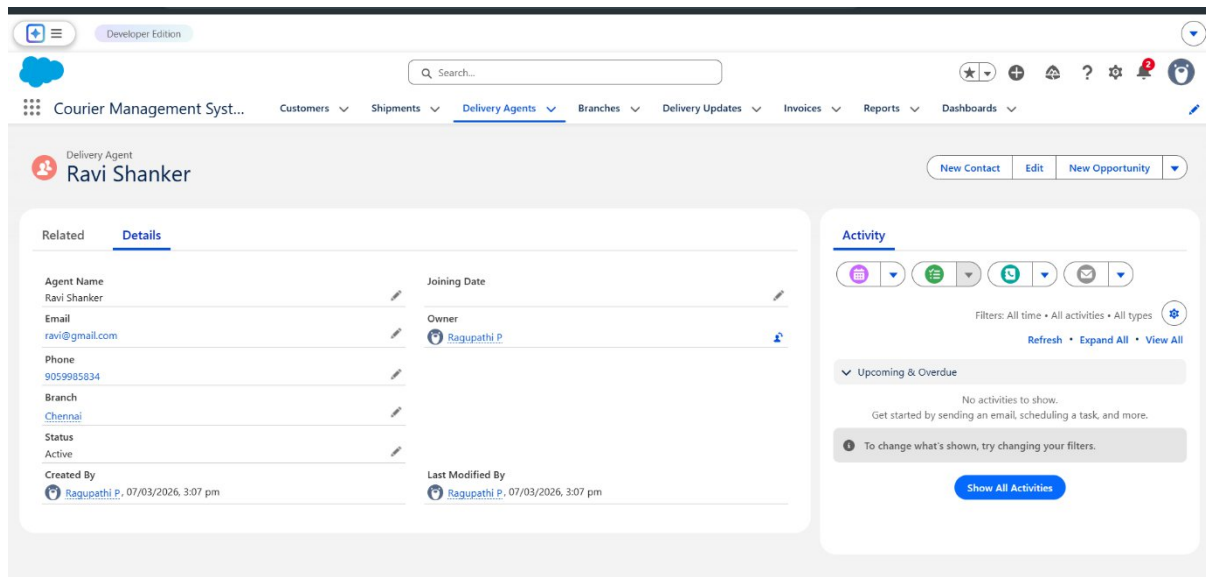
- Stores delivery agent details
- Helps assign agents to shipments
- Tracks delivery responsibilities
- Connects agents with courier branches

Relationship

Object Relationship

Shipment Lookup relationship with Delivery Agent

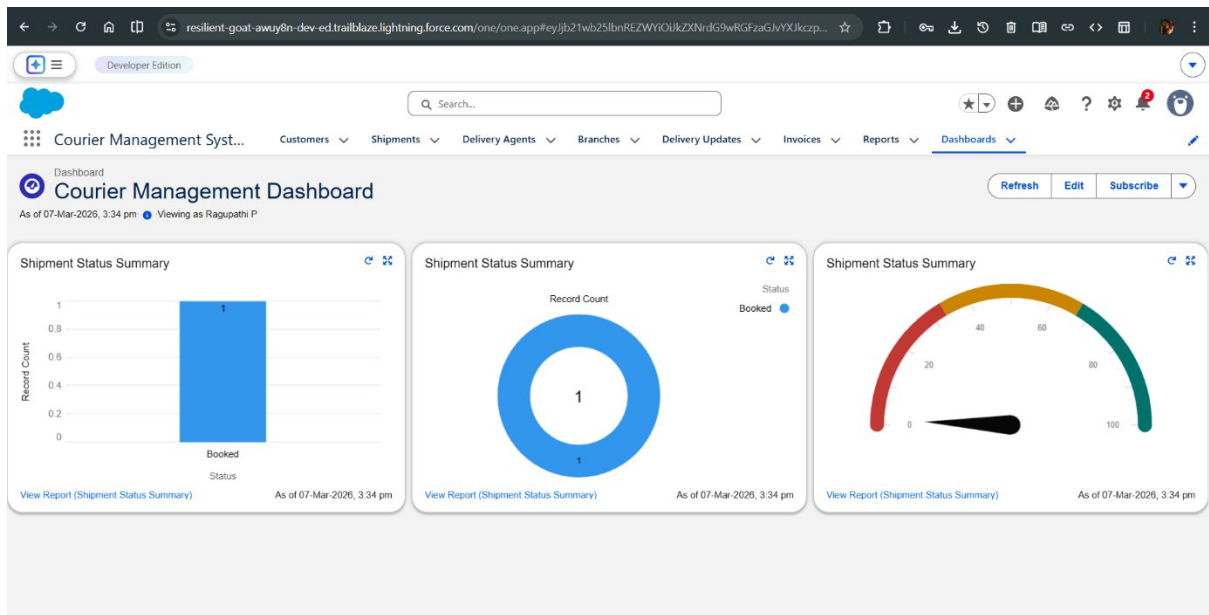
Branch Lookup relationship with Delivery Agent



10. Dashboard in Courier Management System

A **Dashboard in Salesforce** is a visual representation of reports that helps users monitor and analyze important business data. It displays information using charts, graphs, and tables, allowing managers to quickly understand system performance.

In the **Courier Management System**, dashboards are used to track shipment status, delivery performance, and courier revenue. It helps administrators and branch managers monitor courier operations in real time.



Dashboard Components

Component	Description
Shipment Status Chart	Shows shipment stages such as Booked, In Transit, Out for Delivery, and Delivered
Agent Performance Chart	Displays the performance of delivery agents
Revenue Chart	Shows revenue generated from shipments
Delivery Updates	Displays the latest shipment updates

Steps to Create Dashboard

1. Login to **Salesforce**.
2. Click **App Launcher**.
3. Search and select **Dashboards**.
4. Click **New Dashboard**.
5. Enter **Dashboard Name** (Example: Courier Management Dashboard).
6. Click **Create**.
7. Click **Add Component**.
8. Select the report and choose chart type (Bar Chart, Donut Chart, Column Chart).

9. Click **Save**.

The dashboard will display courier system data visually for better decision-making.

11. Explanation

The dashboard helps managers easily analyze courier operations without checking individual records. Visual charts make it easier to understand delivery status, agent performance, and business revenue. This improves decision-making and operational efficiency.

12. Conclusion

The **Courier Management System (CMS)** successfully automates courier operations using the Salesforce platform. The system provides an efficient way to manage customers, shipments, delivery agents, and invoices.

By using automation features such as validation rules, flows, reports, and dashboards, the system reduces manual work and improves operational efficiency. Customers can track their shipments easily, and managers can monitor delivery performance through reports and dashboards.

Overall, the Courier Management System improves courier service quality, increases productivity, and enhances customer satisfaction.